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Assignment-5

Q.1 Objective: Configure Static Route in Cisco packet tracer

• Static route tells the device exactly where to send traffic, no matter what.

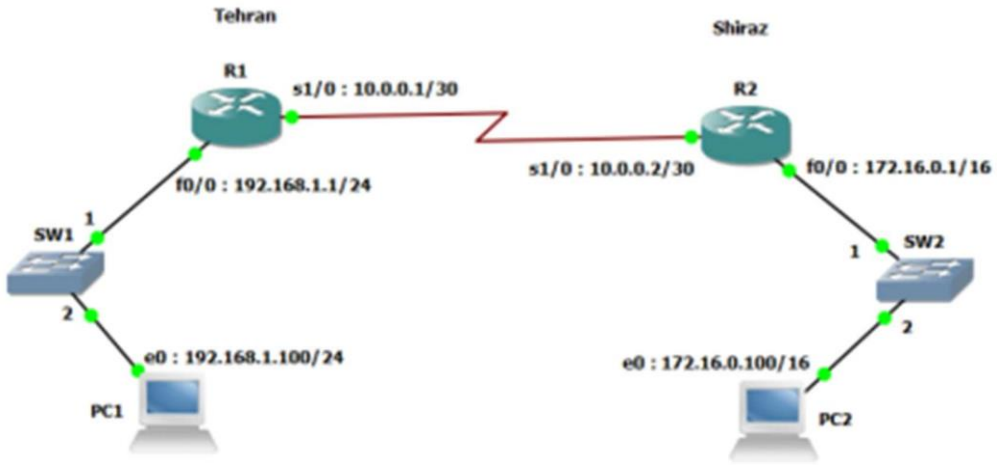
• Static route is often used when your network **has only a few routers** or **there is only one route from a source to a destination**.

Scenario

• Suppose that your company has **2 branches** located in **Tehran** and **Shiraz**. • As the administrator of the network, you are tasked to connect them so that employees in the two LANs **can communicate with each other**.

• After careful consideration you decided to connect them via **static route**.

Physical Topology

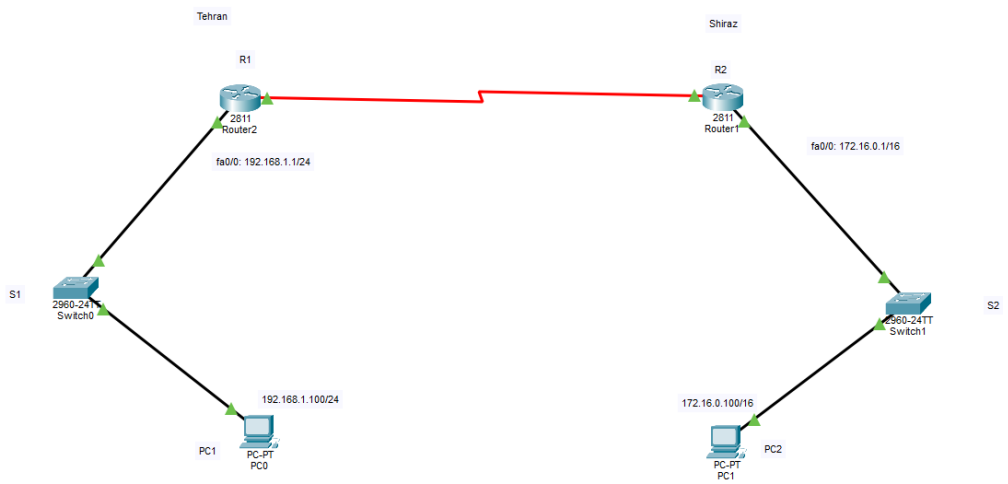


Physical Topology (for Packet Tracer)

Devices required:

- 2 × Routers (e.g., 2811)
- 2 × Switches (2960)
- 2 × PCs
- Serial DCE cable between routers
- Copper straight-through cables for LAN

1. Configure IP address on Router R1 interfaces



2.

Commands:

R1> enable

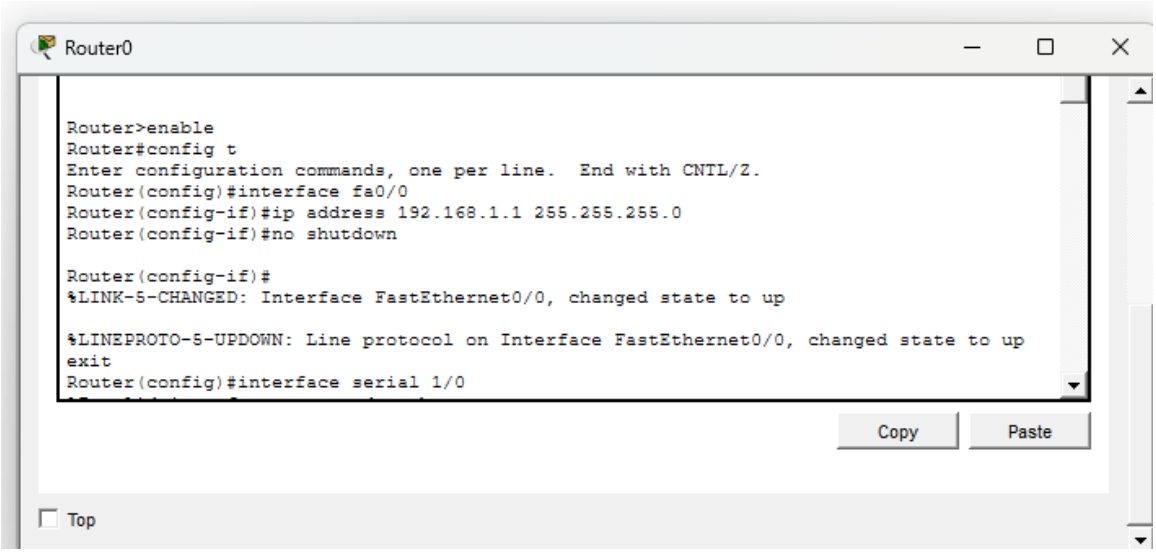
R1# configure terminal

R1(config)# interface fastEthernet 0/0

R1(config-if)# ip address 192.168.1.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)# exit



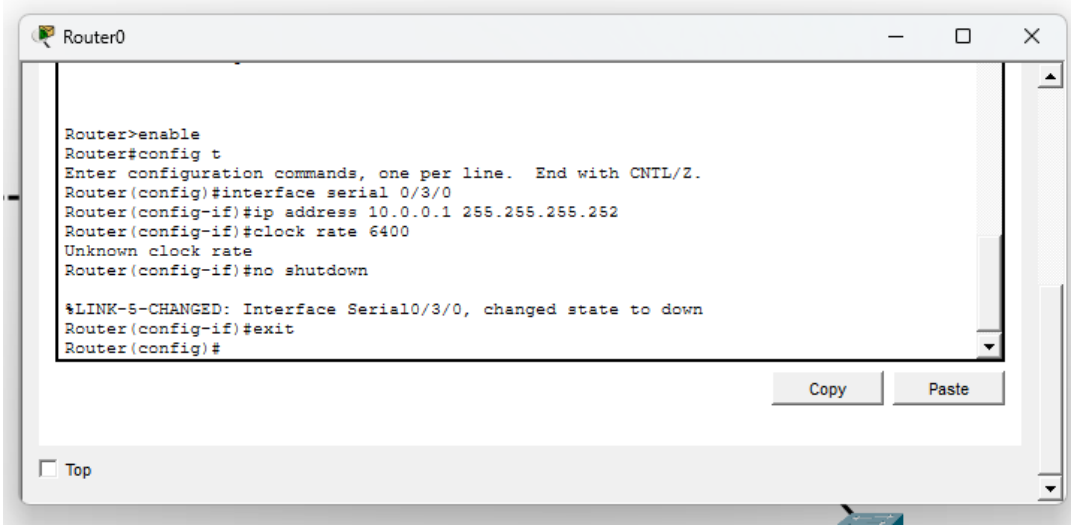
R1(config)# interface serial 1/0

R1(config-if)# ip address 10.0.0.1 255.255.255.252

R1(config-if)# clock rate 64000

R1(config-if)# no shutdown

R1(config-if)# exit



3. Configure IP address on Router R2 interfaces

Commands:

R2> enable

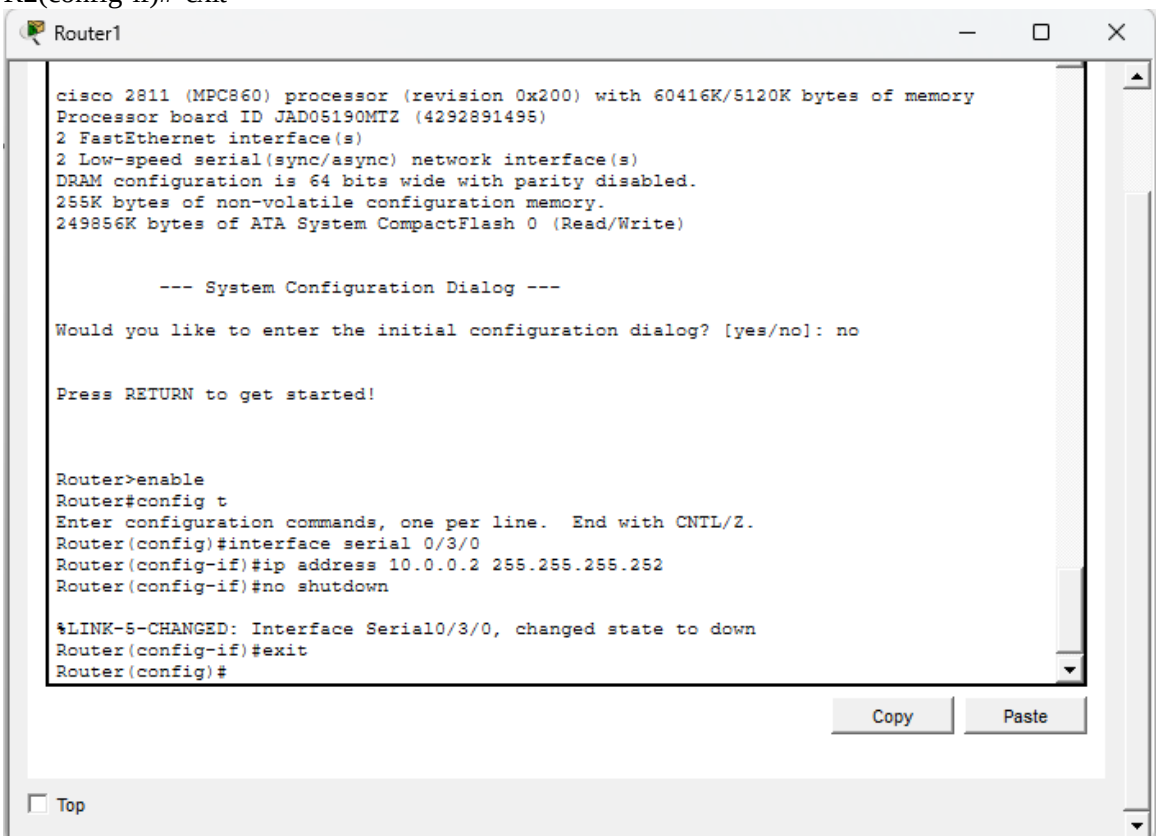
R2# configure terminal

R2(config)# interface serial 1/0

R2(config-if)# ip address 10.0.0.2 255.255.255.252

R2(config-if)# no shutdown

R2(config-if)# exit



R2(config)# interface fastEthernet 0/0

R2(config-if)# ip address 172.16.0.1 255.255.0.0

R2(config-if)# no shutdown

R2(config-if)# exit

```
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

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Top

4. Verify interface configuration

Command (on both routers):

show ip interface brief

Router0

```
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        unassigned      YES unset  administratively down down
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.1        YES manual down                  down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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Top

Router1

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#interface fa0/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.16.0.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual down                  down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

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Top

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5. Display routing table before static routing

Command:

show ip route

Router2

```
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#
```

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Top

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        172.16.0.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Serial0/3/0            10.0.0.2        YES manual down                  down
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/16 is directly connected, FastEthernet0/0
L       172.16.0.1/32 is directly connected, FastEthernet0/0

Router#
```

6. Configure static route on Router R1

(To reach 172.16.0.0 network via R2)

Command:

R1(config)# ip route 172.16.0.0 255.255.0.0 10.0.0.2

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#
```

7. Configure static route on Router R2

(To reach 192.168.1.0 network via R1)

Command:

R2(config)# ip route 192.168.1.0 255.255.255.0 10.0.0.1

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#
```

8. Verify static routes

Command:

show ip route

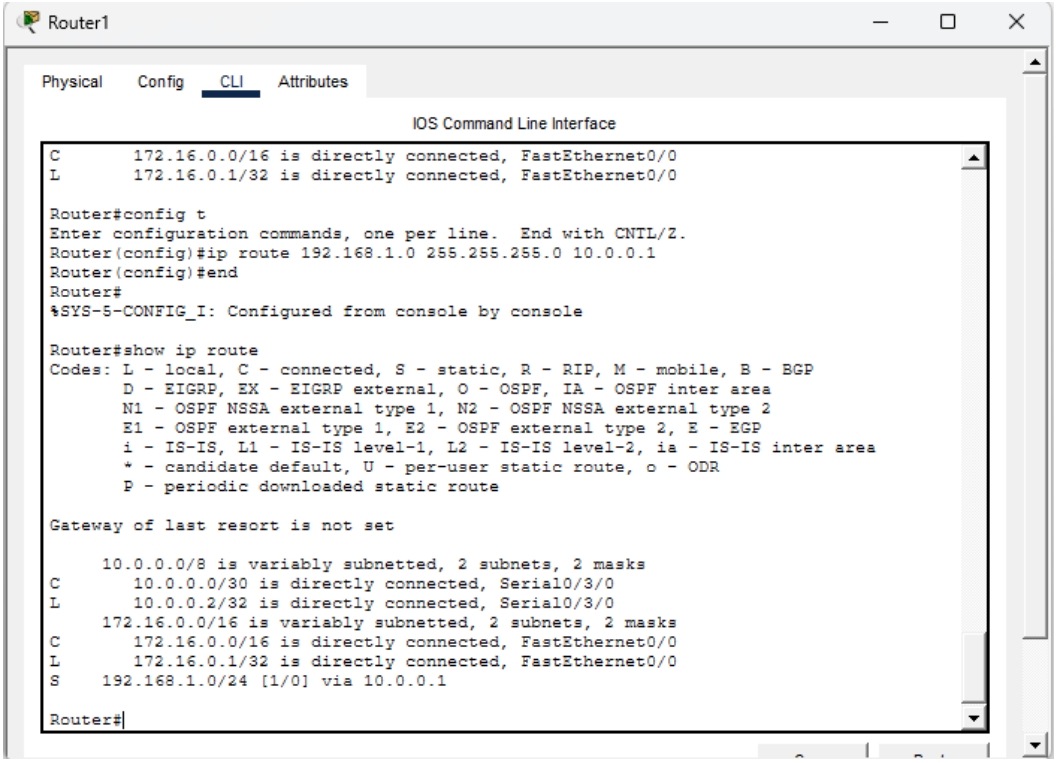
*(Look for routes marked with **S**)*

Router2

Physical Config CLI Attributes

IOS Command Line Interface

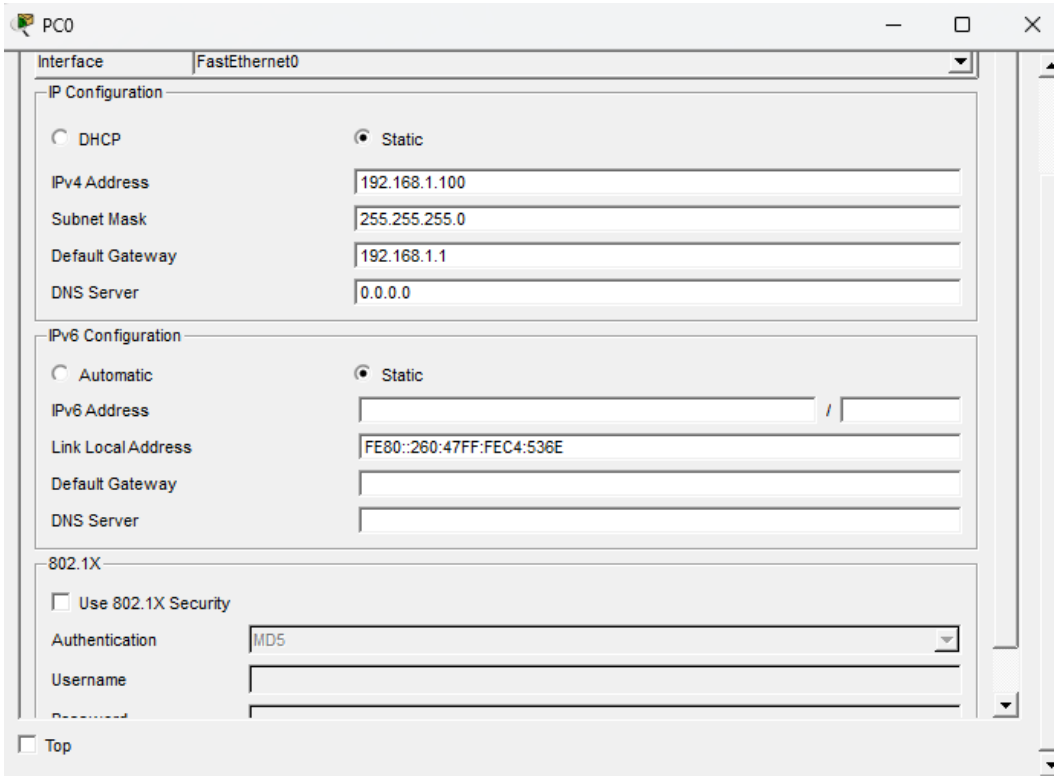
```
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 172.16.0.0 255.255.0.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
S    172.16.0.0/16 [1/0] via 10.0.0.2
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0

Router#
```



9. Configure IP address on PC1

Command (PC Command Prompt):

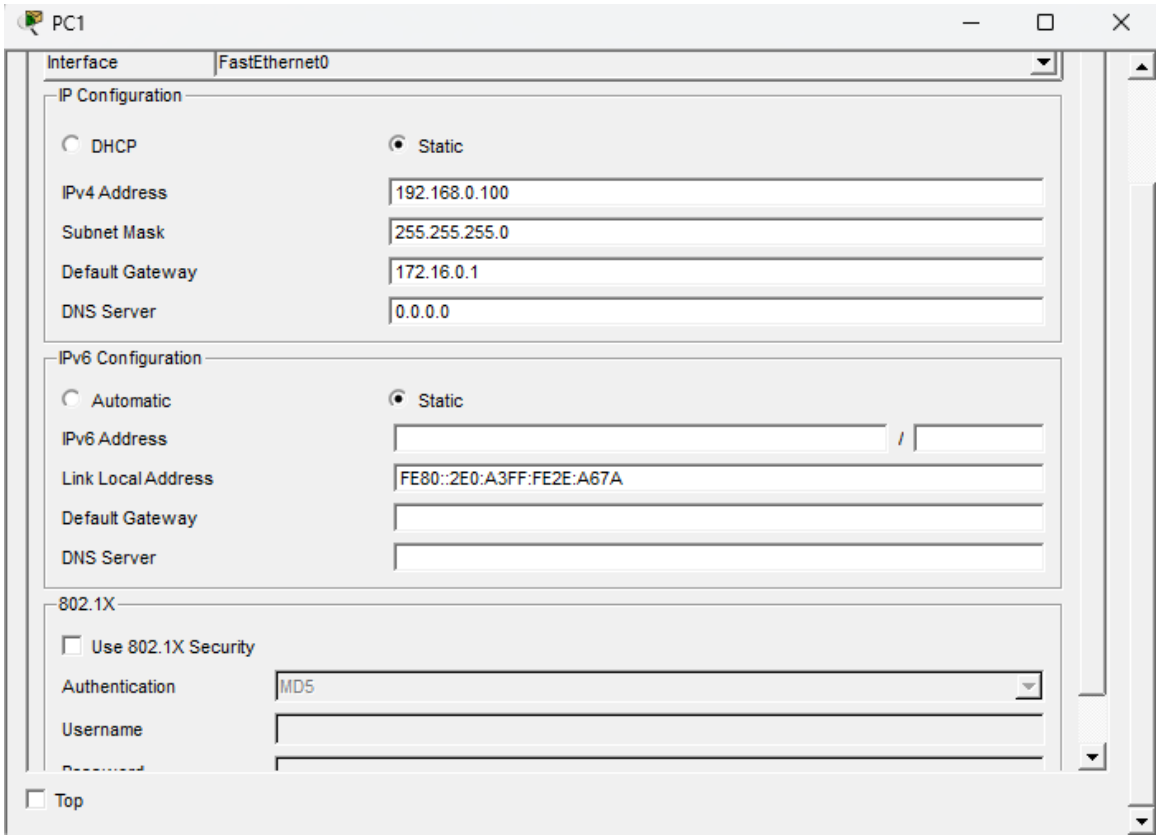
ip 192.168.1.100 255.255.255.0 192.168.1.1



10. Configure IP address on

PC2 Command:

ip 172.16.0.100 255.255.0.0 172.16.0.1



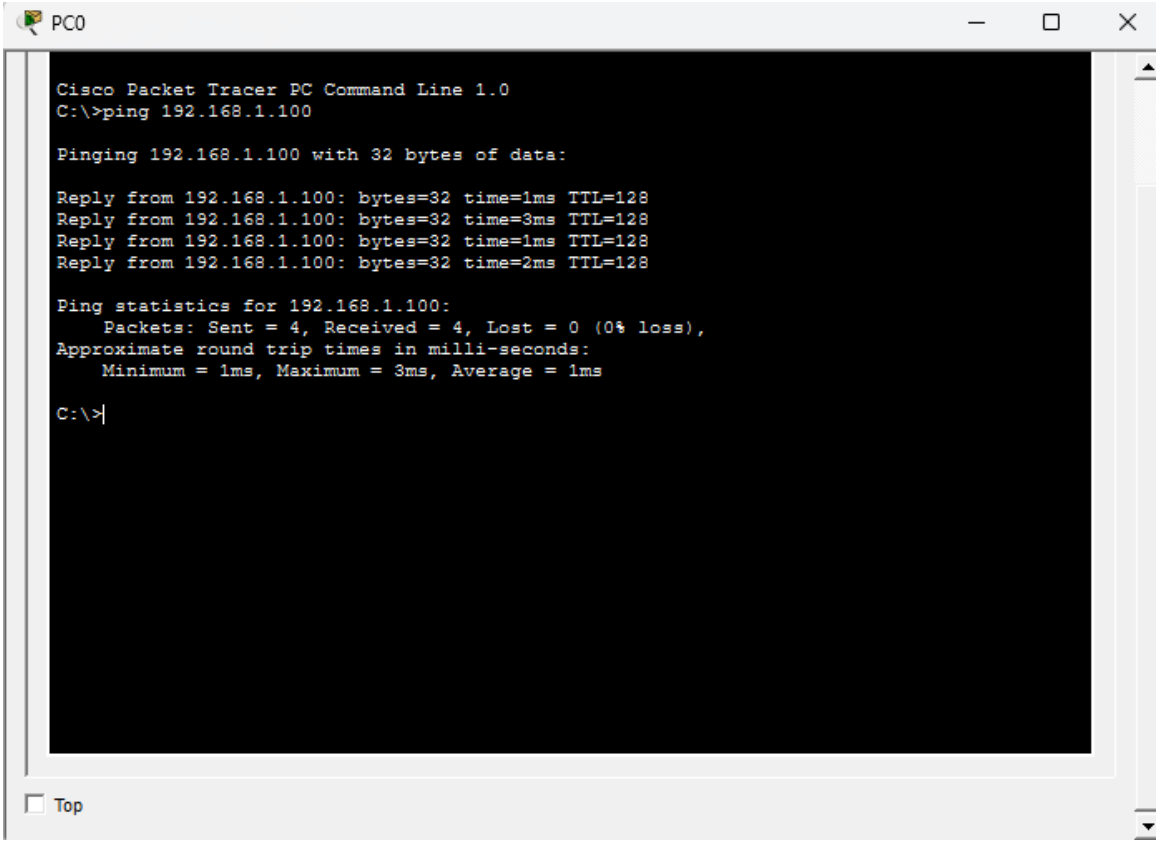
11. Test network connectivity

Command:

ping 192.168.1.100

(or)

ping 172.16.0.100



Result

Successful ping confirms that **static routing is correctly configured** and both networks can communicate.